Integrated Supply Planning as AI Labor

From S&OP through Supply Commit and Allocation Commit — with Powell's Cascaded Framework as the Mathematical Architecture

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1. The Full Planning Cascade

Daybreak's Demand Planning Agent owns the Unconstrained Forecast (UCF). But demand intelligence without supply action is inert. The market requires an integrated solution covering the full cascade from S&OP intent through execution-ready allocation. This memo defines two new agent-owned decisions — the **Supply Commit (SC)** and the **Allocation Commit (AC)** — and describes how Powell's cascaded framework provides the mathematical architecture connecting them to S&OP and MPS/MRP.

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S&OP / IBP Layer          → Planning parameters, priorities, risk appetite
    ↓ feed-forward        ↑ feed-back (outcome-driven parameter re-tuning)
MPS / MRP Layer           → Candidate supply plans from deterministic engines
    ↓ feed-forward        ↑ feed-back (service & cost outcomes re-tune parameters)
Supply Planning Agent     → Supply Commit (SC)
    ↓ feed-forward        ↑ feed-back (allocation shortfalls signal supply gaps)
Allocation Planning Agent → Allocation Commit (AC)
    ↓ feed-forward        ↑ feed-back (execution adherence scores allocation quality)
Execution                 → Order promising, fulfillment, PO release
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Each layer owns a scoped decision, is grounded by mathematical baselines, and is governed by policy and risk thresholds. The layers are connected by **feed-forward contracts** (each layer's output is the next layer's input) and **feed-back signals** (downstream outcomes re-tune upstream parameters). Judgment compounds at every layer rather than resetting each cycle.

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2. Powell's Framework: Feed-Forward and Feed-Back Across Layers

Powell's universal framework models any sequential decision problem with five components: state variables (what we know), decision variables (what we choose), exogenous information (what arrives unpredictably), a transition function (how the system evolves), and an objective function (what we optimize over policies). Each planning layer is a distinct instance of this model, and the architecture's power comes from how instances connect.

Feed-Forward: Contracts Between Layers

Each layer produces an output artifact that becomes a grounding input to the layer below. These are explicit, versioned, hashable contracts — not loose data handoffs:

From	Artifact	To	What It Carries
Demand Agent	UCF (with FBP hash)	Supply Agent	Demand signal, uncertainty quantiles, demand classification
S&OP	Policy envelope	Supply + Allocation Agents	Safety stock targets, OTIF floors, allocation reserves, expedite caps
MRP Engine	SupBP candidates	Supply Agent	Multiple parameterized supply plans on the service-vs-cost frontier
Supply Agent	Supply Commit (with SupBP hash)	Allocation Agent	Available supply by item-location-bucket, supply pegging
Allocation Solver	SBP candidates	Allocation Agent	Multiple allocation proposals with constraint diagnostics and marginal values
Allocation Agent	Allocation Commit (with SBP hash)	Execution	Entitlement caps and priority ordering by item-segment-bucket

Every published decision references the hash of its grounding inputs. This preserves decision identity and makes the entire cascade auditable and reproducible — a structural requirement of AI-as-Labor.

Feed-Back: Outcome Signals That Re-Tune Upstream Policies

Powell's base-model/lookahead-model distinction creates the feedback architecture. The **base model** is a stochastic simulation of the full cascade; the **lookahead models** (MRP, allocation solver) are approximate models solved at each decision point. An optimal solution to the lookahead model is not an optimal policy — the quality depends on parameterization. Feed-back is the mechanism by which parameterization improves.

Outcome Signal	Measured At	Fed Back To	What It Re-Tunes
Actual OTIF by segment	Execution	Supply Agent + S&OP	Safety stock targets, lead time buffers, demand inflation factors
Allocation shortfalls (segment shorted despite available supply)	Allocation Agent	Supply Agent	Source mix, timing, buffer allocation by criticality
Execution adherence (AC vs actual fulfillment)	Execution	Allocation Agent	Priority ordering, entitlement calibration, trust collapse detection
Expedite frequency and cost	Execution	Supply Agent + S&OP	Lead time buffers, supplier diversification targets, expedite budget
E&O write-offs	Inventory	Supply Agent + S&OP	DOS ceilings, forward buy authorization, lot size parameters
Override outcome delta (did the human override improve results?)	All layers	Agent learning loop	Override absorption or rejection; policy/parameter adjustment

The override feed-back is Daybreak's structural moat. At every layer, human overrides are captured with full context, tied to a persistent decision identity, and scored against the agent baseline. Overrides that consistently improve outcomes are absorbed; overrides that destroy value are flagged. This is Powell's compounding loop mapped to Daybreak's judgment layer: **the system learns from the decisions that matter, not from historical data alone.**

The critical cross-layer feed-back is **allocation shortfall → supply re-tuning.** When the Allocation Agent repeatedly shorts a segment because insufficient supply was ordered, that signal propagates to the Supply Agent (order more, buffer differently, diversify sources) and to S&OP (raise the safety stock target for that category). Without this cross-layer signal, the Supply Agent optimizes in isolation and the Allocation Agent is left rationing a bad hand. Powell's base model simulation makes this visible because it evaluates the full cascade, not individual layers.

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3. The Supply Planning Agent: Supply Commit (SC)

The **Supply Commit** is the agent-owned decision: the best demand-informed, constraint-respecting set of supply actions (POs, production orders, transfer orders) that balances inventory investment against service risk within the S&OP policy envelope.

Decision artifact: order_qty [item, source, destination, time_bucket], target_inventory_position, projected_otif [item, segment, bucket], projected_inventory_cost, supply_pegging [demand → supply action].

Ownership boundary: The agent owns applying policies and constraints, explaining decisions through pegging and attribution, recommending source selection, and detecting unsafe auto-submissions. It does **not** own the UCF, supplier contracts, capacity expansion, allocation to segments (Allocation Agent), S&OP consensus, or financial targets as decision logic.

Supply Baseline Pack (SupBP)

Non-authoritative mathematical grounding. The agent may accept, reject, or blend candidates.

Contents: (1) **Candidate supply plans** — multiple MRP runs with different parameterizations (standard, service-maximized, min-cost, simulation-optimized CFA, scenario-hedged). Each includes order quantities, projected inventory, projected OTIF, projected cost. (2) **Demand grounding** — UCF hash, demand by item-location-bucket, uncertainty quantiles, demand classification. (3) **Supply-side constraints** — supplier capacity, lead time distributions, MOQ/lot sizes, reliability scores, cost structures, open orders, substitution eligibility. (4) **Inventory health** — on-hand, in-transit, committed, available-to-plan, DOS, E&O exposure, shelf-life. (5) **Tradeoff signals** — marginal OTIF impact per unit of safety stock, cost delta between sources, lead time sensitivity, stockout probability. (6) **Versioning** — all hashes for reproducibility.

Supply Policies

Integrity (block submission): Inventory Balance (negative projected inventory), Lead Time Feasibility (order cannot arrive in time), Demand Coverage (unaddressed demand with no documented reason). **Outcome risk (mark Suggested):** Service Risk (OTIF below floor), Inventory Investment (DOS exceeds ceiling), Supplier Concentration (single-source share too high), Expedite Exposure, E&O Exposure, Plan Stability (period-over-period whiplash).

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4. The Allocation Planning Agent: Allocation Commit (AC)

The **Allocation Commit** is the agent-owned decision: the best constraint-respecting distribution of constrained supply across demand segments and time buckets, expressed as entitlement caps and priority ordering for execution.

Decision artifact: entitlement_qty [item, segment, time_bucket], allocation_priority, unallocated_qty, pegging_summary [supply_pool → segment attribution + gating factors].

Ownership boundary: The agent owns applying the policy bundle to produce the AC, explaining through pegging, and detecting unsafe auto-submissions. It does **not** own segmentation definitions, customer tiering, contractual policy, creating supply (Supply Agent), demand shaping, S&OP reconciliation, or order-level promising.

Solver Baseline Pack (SBP)

Non-authoritative. Candidates include fair-share, priority heuristic, LP, MILP, and hybrid outputs.

Contents: (1) **Candidate Allocation Commits —** entitlement quantities, priorities, unallocated quantities, pegging maps. (2) **Feasibility and constraint diagnostics —** binding constraints, slack, gating factors, infeasibility reasons. (3) **Marginal value signals —** shadow prices for supply/capacity, marginal short penalty by segment (Powell's VFA outputs: where one more unit matters most). (4) **Policy compliance metrics —** service floor breaches, fair-share deviations, priority inversions, stability deltas. (5) **Versioning —** all hashes for reproducibility.

Allocation Policies

Integrity (block submission): Supply Conservation (allocated > available), Temporal Feasibility (allocation before availability), Eligibility & Compliance (regulatory violations), Pegging Explainability (unattributable shortfalls). **Outcome risk (mark Suggested):** Service Floor Compliance, Fair-Share Deviation, Priority Inversion, Allocation Stability, Expiry/Obsolescence Exposure.

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5. S&OP and MPS/MRP: Selling Together or Separately

When Sold as the Integrated System

S&OP operates as the **parametric CFA envelope —** it produces the tunable parameters governing downstream agents. MPS/MRP operates as the **deterministic lookahead engine —** it generates candidate supply plans by running with multiple parameterizations. Daybreak's base model simulation quantifies S&OP tradeoffs ("if we tighten the OTIF floor from 95% to 98%, inventory investment increases $2.1M and expedite cost drops $400K") and optimizes MRP parameters through the Powell CFA tuning loop.

When Supply + Allocation Agents Are Sold Without S&OP/MPS Layers

The agents remain fully functional. The customer simply provides the inputs that S&OP and MPS would otherwise generate:

Normally Produced By	What the Customer Must Provide	Format	Frequency
S&OP — Safety stock targets	Weeks-of-supply target by item or category	CSV/API: [item_or_category, target_wos]	Monthly or quarterly
S&OP — OTIF floors	Minimum service level by customer tier or segment	CSV/API: [segment, otif_floor_pct]	Monthly or quarterly
S&OP — Allocation reserves	% of constrained supply reserved for priority segments	CSV/API: [segment, reserve_pct]	Monthly or quarterly
S&OP — Expedite/budget caps	Maximum expedite spend by category	CSV/API: [category, max_expedite_$]	Monthly or quarterly
MPS/MRP — Candidate supply plans	At minimum one supply plan (their existing MRP output)	CSV/API: standard PO/TO format with [item, source, dest, qty, date]	Per planning cycle
MPS/MRP — BOM and lead times	Bill of materials, supplier lead times, MOQs, lot sizes	CSV/API: master data format	On change

What changes without these layers:

- **No S&OP simulation layer:** The customer loses the ability to run "what-if" scenarios across the full cascade. The agents still enforce the policy parameters provided, but the parameters are tuned by the customer's existing S&OP process rather than by Daybreak's simulation engine. Feed-back still operates — outcome signals still surface — but the parameter re-tuning loop is manual rather than simulation-assisted.
- **No MRP candidate generation:** The SupBP contains only the customer's single MRP output as a candidate rather than multiple parameterized candidates on the tradeoff frontier. The Supply Agent still reasons about the plan (applying policies, checking constraints, flagging risks), but cannot show the service-vs-cost tradeoff surface. The agent's value shifts from "select the best candidate" to "validate and govern the customer's plan."
- **The upgrade path is clear:** When the customer is ready, adding the MPS/MRP layer means Daybreak generates multiple SupBP candidates and optimizes parameters via simulation. Adding the S&OP layer means Daybreak quantifies the consequences of policy changes and tightens the feed-back loop. Each layer adds incremental value without requiring re-architecture.

When MPS/MRP Is Sold Without S&OP

The customer provides S&OP parameters (safety stock targets, OTIF floors, reserves, caps) as static inputs.

Daybreak's MRP engine generates parameterized candidates, the Supply Agent selects, and the Allocation Agent distributes. The feed-back loop operates within the MRP-and-below layers (MRP parameters are tuned against outcomes), but S&OP parameter re-tuning remains the customer's responsibility.

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6. UX Primitives and Metrics

Consistent UX Across Both Agents

Primitive	Supply Planner	Allocation Planner
Scheduled Tasks	"847 supply decisions committed. 23 need attention before 2pm PO cutoff. Top risk: gloves at 88% OTIF."	"312 allocations committed. 8 flagged. Top risk: wound care — strategic hospitals shorted 15%."
Worklist	Exceptions by policy type, sorted by $ exposure. Each: SC vs SupBP candidates, OTIF, DOS, gating constraint.	Exceptions by policy type. Each: AC vs SBP candidates, service floor status, fair-share deviation, pegging.
Chat	"Why 40% more from Supplier B?" → reliability drop, sensitivity analysis, cost delta. Planner injects: "Port delays through March — extend buffer."	"Which customers shorted?" → pegging to constraint. "Reserve 500 for Hospital Alpha — CEO commitment." → captured with context.
Task Log	Audit: candidates generated, agent rationale, policies evaluated, auto-submissions.	Audit: candidates generated, agent rationale, policies evaluated, auto-submissions.
Agent Config	OTIF floors, DOS ceilings, expedite caps, supplier parameters, source eligibility.	Service floors, fair-share tolerance, priority rules, segment definitions, entitlement rules.
Dashboards	Inventory health, supplier trends, service-vs-investment Pareto, stockout heatmap.	Coverage by segment, floor compliance, fair-share tracking, priority inversion log.

Decision Status Paths (Both Agents)

Suggested → Reviewed → Accepted → Submitted | Suggested → Reviewed → Overridden → Submitted | Autonomous → (Reviewed → Accepted/Overridden →) Submitted | Autonomous → Not Reviewed → Auto-Submitted

Key Metrics

Metric	Definition (both agents)
Touchless Rate	% of decisions not reviewed

Metric	Definition (both agents)
Agent DQS	Outcome lift from agent decisions vs baseline
User DQS	Outcome lift from human overrides vs agent baseline
Cognitive Load Index	Decisions reviewed ÷ total in scope
Override Dependency (Kill Metric)	Repeated intervention on same decision identity across cycles
Downstream Coherence	Supply: SC vs what Allocation receives. Allocation: AC vs what execution fulfills.

If either agent requires persistent human intervention, it is **not** AI-as-Labor.

Summary

Daybreak's market offering is a **cascaded planning system where judgment compounds at every layer.** Powell's framework provides the mathematical guarantee: S&OP sets parametric policies → MRP generates parameterized candidates → Supply Agent owns the Supply Commit → Allocation Agent owns the Allocation Commit → execution outcomes feed back to re-tune every layer. The feed-forward contracts ensure each layer has the grounding it needs. The feed-back signals ensure the system self-corrects. The modular architecture ensures each layer can be sold independently, with clear input contracts when upstream layers are absent, and a natural upgrade path when the customer is ready to compound judgment across the full cascade.

Legacy systems discard decision lineage every cycle. They store values, not judgment. Daybreak stores judgment as a first-class system object at every layer. The company that compounds judgment fastest wins.